



Global Malaria Programme

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Digital solutions for malaria surveillance: DHIS2 and NMDR

DHIS2

Surveillance is a core malaria intervention. To help countries strengthen routine surveillance systems and promote data use for decision-making across all transmission settings, WHO has developed a suite of digital tools, standards and training materials.

These tools are implemented through malaria modules in DHIS2, which reflect WHO recommendations and best practices. Countries already using DHIS2 – or planning to adopt it – can apply these tools directly. Others can use the modules as guidance to develop or enhance their own surveillance systems.

The [DHIS2 malaria modules](#) include 3 main branches with multiple components (see boxes below for more information on the routine and entomology modules).

DHIS2 malaria modules

[View the modules](#)

Routine

Aggregate module + Case-based module

→ Supports routine reporting and analysis of service delivery, treatment and other activities, with datasets and indicators tailored to both burden reduction and elimination settings.

Entomology and vector control

Entomological surveillance & vector control interventions

→ Supports countries in improving the collection and use of entomological and vector control data to inform programmatic decisions.

Additional modules/dashboards

- Seasonal malaria chemoprevention (SMC)
- Facility stock monitoring and triangulation
- Malaria vaccine module
- Data quality dashboard
- District dashboard

Each module provides a standardized set of data elements, indicators, validation rules and dashboards for visualizing core epidemiological and data quality metrics. The visualizations – such as charts, tables, and maps – enable rapid dissemination of information to decision-makers. The modules are configurable and can be used independently or in combination, depending on the public health response required. They are supported by comprehensive guidance and training materials to facilitate implementation and use at facility and programme levels.

Aggregate module

In high transmission settings, where the goal is to reduce malaria-related morbidity and mortality, surveillance data are aggregated to provide a broad picture of disease burden. This includes:

- **Monitoring trends in cases and deaths over time**
- **Mapping geographic distribution of malaria**
- **Analysing demographic characteristics of affected populations**
- **Understanding seasonal transmission patterns**

Such data support stratification of geographic units by malaria prevalence or annual parasite incidence, enabling targeted interventions. They also inform planning for essential commodities such as test kits and antimalarial medicines.

The aggregate module is aligned with the [reference manual on malaria surveillance, monitoring and evaluation](#), and the [framework for malaria elimination](#), and has been adopted by over 40 countries. Installations range from full implementations with standard dashboards (e.g. Data Quality, LMIS, District Trends) to customized variants tailored to national needs.

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Case-based malaria module

In elimination settings, early detection and rapid response are critical. The case-based module, which will be released soon, supports a range of functions, including:

- **Line listing of suspected cases (optional)**
- **Diagnosis and treatment**
- **Treatment follow-up (optional)**
- **Case investigation**
- **Foci investigation and response**
- **Follow-up activities**

Data collected through this module feed into elimination dashboards for analysis and

reporting, enabling programmes to identify transmission drivers and implement targeted responses.

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Entomological surveillance and vector control interventions

The modules include a set of standard data collection forms, automatically calculated indicators, data visualizations, and thematic dashboards that enable users to collect, visualize, and interpret data from the following activities, in line with WHO recommendations:

- ITN bioefficacy monitoring
- IRS residual efficacy monitoring
- Larviciding
- IRS mass campaigns
- ITN mass campaigns

See also [this page](#) for more information.

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National malaria data repositories (NMDR)

The National Malaria Data Repository (NMDR) is a country-owned digital platform that brings together malaria-related data from multiple sources into a single, structured system. It improves the availability, accessibility and use of high-quality data for decision-making at all levels of the health system.

An NMDR serves as a centralized location to store, manage and access malaria-related data across domains such as epidemiology, entomology, vector control, chemoprevention and finance. It also includes contextual information, such as population denominators. The NMDR is typically structured as a data warehouse or integrated information system, and supports national malaria programmes by organizing and visualizing this information through indicators and dashboards.



7 OCTOBER 2025

[Guidance on establishing a national malaria data repository](#)

Version 1.0, July 2025

[Download](#)

[Read More](#)

Rather than replacing existing systems, the NMDR acts as a unifying layer that connects and harmonizes data from platforms such as DHIS2 and LMIS. It includes tools for analysis, mapping, and visualization, and is a key component of modern malaria surveillance architecture.

NMDR guidance

The initial version of the NMDR guidance focuses on implementation using DHIS2. It outlines a phased approach to help countries plan, build, and operationalize their NMDR in alignment with WHO recommendations and national priorities.

Development phases

Developing and institutionalizing an NMDR typically takes several years, involving multiple iterations and the addition of modules. However, an NMDR can begin adding value within the first year, once the first module is completed.

Broadly, NMDR development follows 4 phases:

1. Planning the NMDR

- Establish the vision, scope and objectives. This includes conducting a readiness assessment, engaging stakeholders, and identifying existing systems and data sources.
- 2. Initial NMDR setup and data integration
 - Configure the platform, integrate initial datasets (often starting with routine health facility data) and establish basic dashboards and metadata.
- 3. Intermediate NMDR development
 - Expand through iterative integration of additional modules, such as entomology, vector control, stock monitoring and case-based surveillance. This phase includes training, roll-out and early use of the system.
- 4. Advanced NMDR development
 - Focus on governance, sustainability and continued improvement. This includes supervision, refinement of data flows and institutionalization of the NMDR within national systems.

Phases 3 and 4 are repeated as modules are progressively prioritized, installed, customized, rolled out and institutionalized.

Readiness assessment

A readiness assessment is a critical first step. It helps countries understand their current digital landscape, identify gaps and determine the feasibility and scope of NMDR implementation. This includes reviewing existing surveillance platforms, data standards, human resources and governance structures.

Annexes included in the guidance

The NMDR guidance includes several annexes that provide practical tools and reference materials

- **Workplan and budget templates**
- **Readiness assessment**
- **Modules listing**
- **Master indicator listing**

Implementation support tools

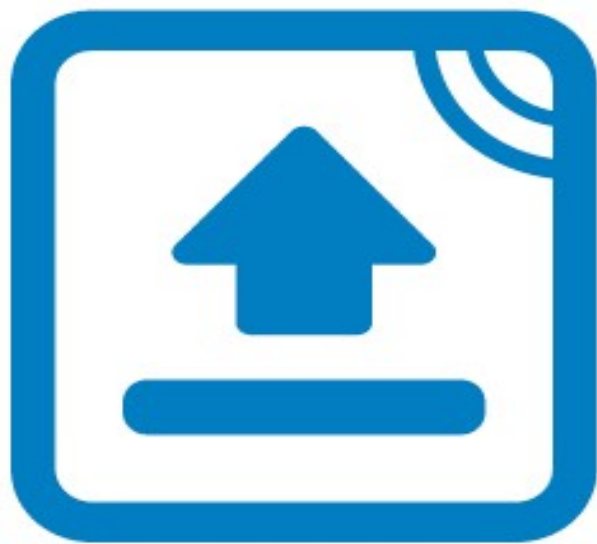
WHO is working in collaboration with partners to support to countries implement the modules. To facilitate implementation, WHO has developed multiple tools:

- to customize the modules to country needs (D2-docker);
- to facilitate installation of the modules (MetaData Sync App);
- to help countries build capacity for the use of the modules (Training App);
- to facilitate collation, sharing and reporting of data (MetaData Sync); and
- to facilitate import of data from excel sheets (Bulk Load).



Metadata Sync DHIS2 app

A DHIS2 application that allows for the exchange of data and metadata between different DHIS2 instances and for the generation, exploration and import of DHIS2 standard packages.



Bulk Load

A DHSI2 application to facilitate data import from excel. This tool helps countries and their implementing partners to bring historical and prospective data collected with excel into DHIS2.



D2-docker

A tool to facilitate the development of metadata. This tool helps countries and their implementing partners to adapt WHO standard metadata to country needs.



GEE Google Earth Engine App and Data Importer

This application allows countries to bring climate and environment data into DHIS2 to better understand the dynamics of malaria transmission.



DHIS2 interactive end-user tutorial

This tutorial walks users through different functions of DHIS2, in an interactive manner, to help them accomplish the tasks of reporting data, generating visualizations dashboards and importing data from MS Excel.



DHIS2 standard packages demo

Explore the WHO standard metadata packages for entomology and vector control and supporting DHIS2 application in our life demo. Available in English, French, Portuguese and Spanish.

Additional information

Infographic

Overview of DHIS2 malaria modules implementation

20 May 2021

Improving malaria surveillance using DHIS2 and the WHO malaria module